GOOGLE CLOUD HANDS-ON LAB

Using Cloud Shell and the gcloud Command-Line Tool

Step-by-step practical documentation with screenshot placement guide

Prepared from the supplied lab instructions

1. Overview

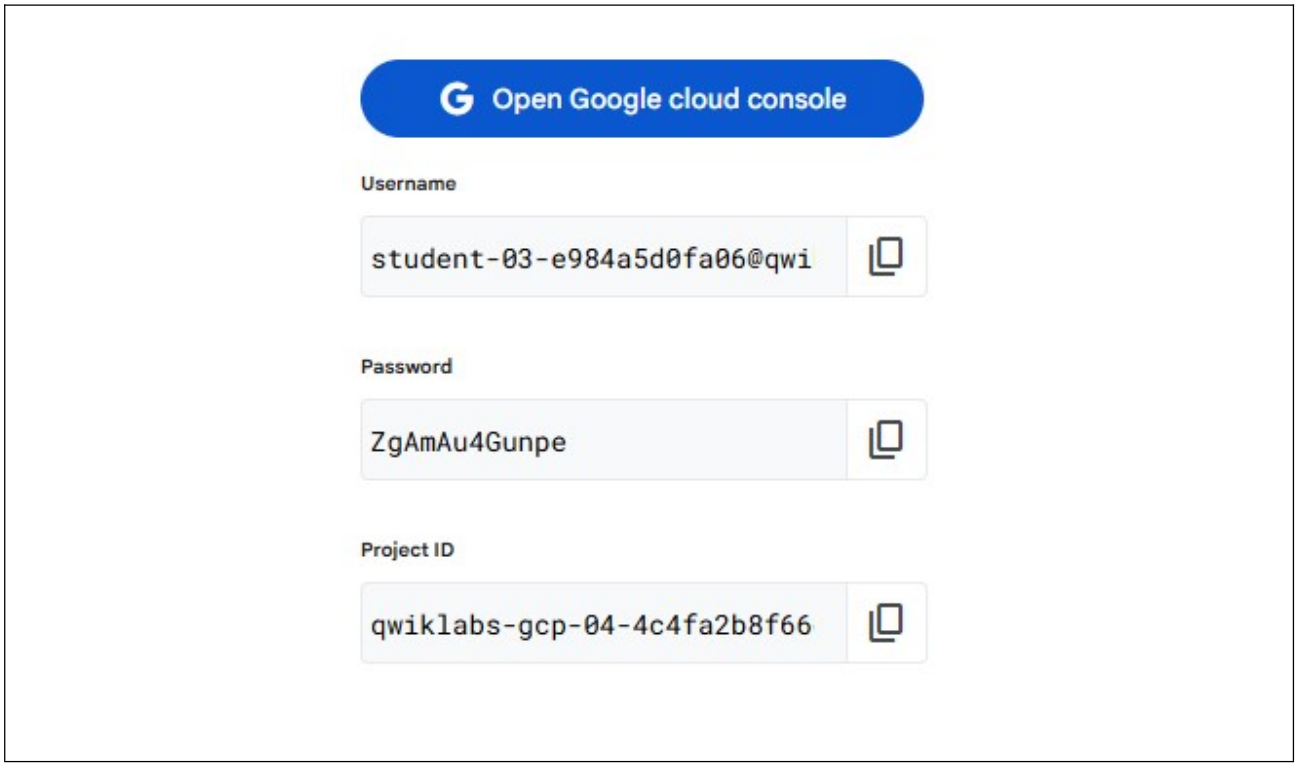
Google Cloud Shell provides command-line access to Google Cloud resources. It runs in a browser, includes the gcloud command-line tool, and provides a persistent home directory. In this lab, we configure the environment, create and connect to a Compute Engine virtual machine (VM), filter command output, configure HTTP access, and inspect Cloud Logging logs.

2. Objectives

- Use common gcloud commands.
- Set the default Compute Engine region and zone.
- Create a VM instance named gcelab2.
- Filter VM and firewall-rule output.
- Connect to the VM using SSH and install Nginx.
- Configure a firewall rule for HTTP traffic.
- View logs related to Compute Engine resources.

3. Requirements and safety notes

- Use a standard web browser; Chrome is recommended.
- Use an Incognito/private window to avoid mixing your personal account with the temporary lab account.
- Use only the temporary credentials supplied by the lab.
- Do not sign up for a free trial or add recovery options to the temporary lab account.
- The lab timer starts when you click Start Lab and the lab cannot be paused.



4. Start the lab and sign in

Step 1: Start the lab

Open the lab instructions and click Start Lab. Wait for the temporary credentials and Open Google Cloud console button to appear.

Step 2: Open Google Cloud Console

Click Open Google Cloud console. If prompted to choose an account, select Use another account.

Step 3: Sign in with the temporary account

Copy the username and password from the lab access panel. Enter the username, click Next, enter the password, and continue.

Step 4: Complete first-time prompts

Accept the terms if prompted. Skip recovery options, two-factor authentication, and free-trial registration for this temporary account.

5. Activate Cloud Shell

Step 5: Open Cloud Shell

In the Google Cloud Console, click the Activate Cloud Shell icon near the top of the page.

Step 6: Authorize Cloud Shell

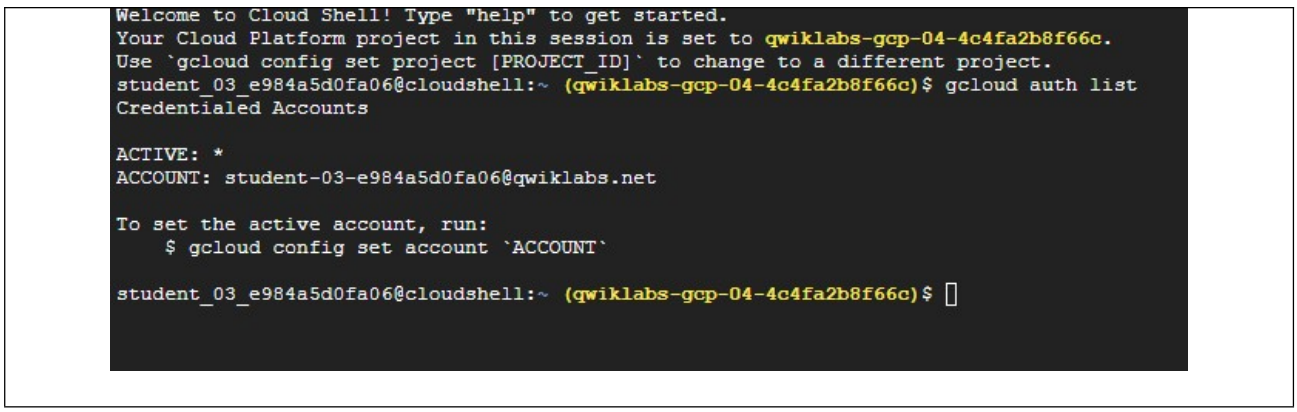
Continue through the information prompt and click Authorize when asked. Wait for the terminal to connect to the active project.

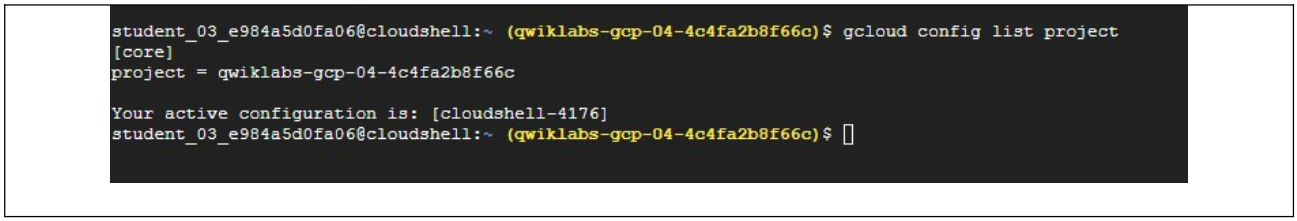
Optional verification commands:

```
gcloud auth list

gcloud config list project
```

The active account is marked with an asterisk in gcloud auth list. The project command shows the current project ID.





6. Task 1 — Configure the environment

Step 7: Understand regions and zones

A region is a geographic area. A zone is a deployment area within a region. VM instances and persistent disks are zonal resources, so resources that must be attached together need compatible locations.

Step 8: Set the region

Replace REGION with the region specified by your lab or the value you intend to use.

```
gcloud config set compute/region REGION

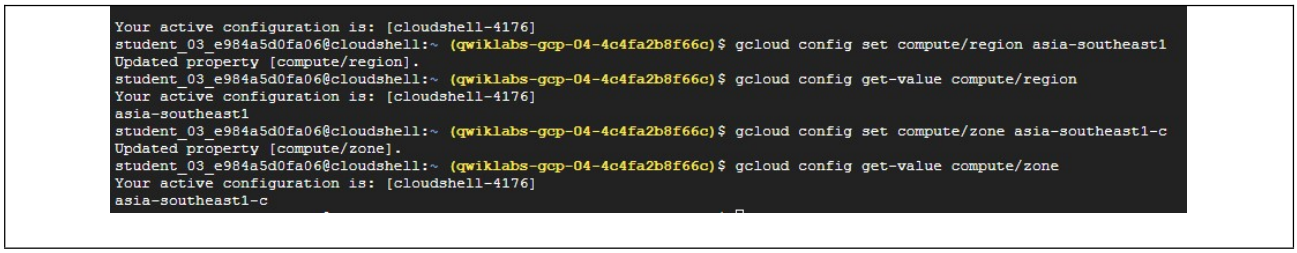
gcloud config get-value compute/region
```

Step 9: Set the zone

Replace ZONE with the zone specified by your lab. Make sure the zone belongs to the selected region.

```
gcloud config set compute/zone ZONE

gcloud config get-value compute/zone
```



Step 10: Check project information

```
gcloud config get-value project

gcloud compute project-info describe --project $(gcloud config get-value project)
```

Review the output for the project ID and any default region/zone metadata. The exact output depends on the temporary lab project.

7. Create environment variables

Store the project ID and zone in shell variables to make later commands shorter.

```
export PROJECT_ID=$(gcloud config get-value project)

export ZONE=$(gcloud config get-value compute/zone)

echo -e "PROJECT ID: $PROJECT_ID\nZONE: $ZONE"
```

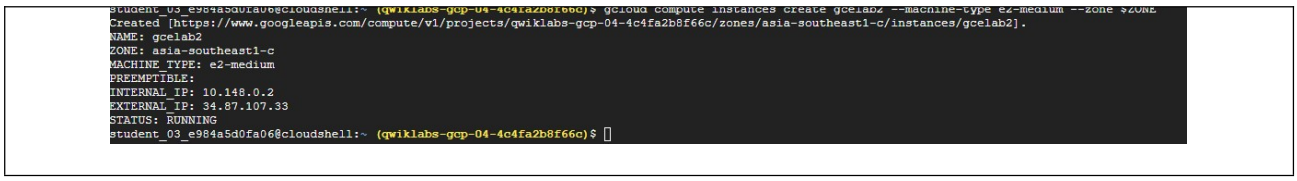


8. Create a Compute Engine VM

Create a virtual machine named gcelab2 using the e2-medium machine type in the selected zone.

```
gcloud compute instances create gcelab2 --machine-type e2-medium --zone $ZONE
```

Wait for the command to finish. The output should show the VM name, zone, machine type, IP addresses, and status. Confirm that the status is RUNNING.



In the lab page, click Check my progress for the objective “Create a virtual machine with gcloud.”

Step 11: Explore command help

```
gcloud compute instances create --help
```

Press Q to exit the help viewer. You can also use these commands to explore gcloud help:

```
gcloud -h

gcloud config --help

gcloud help config

gcloud config list

gcloud config list --all

gcloud components list
```

```
wstudent_03_e984a5d0fa06@cloudshell:~ (qwiklabs-gcp-04-4c4fa2b8f66c)$ gcloud -h
Usage: gcloud [optional flags] <group | command>
  group may be            access-approval | access-context-manager |
                          active-directory | agent-identity | agent-registry |
                          ai | ai-platform | alloydb | alpha | anthos |
                          api-gateway | apigee | apihub | app | app-topology |
                          apphub | artifacts | asset | assured | audit-manager |
                          auth | backup-dr | batch | beta | beyondcorp |
                          biglake | bigtable | billing | bms | bq | builds |
                          certificate-manager | cloud-shell |
                          cloudlocationfinder | cluster-director | colab |
                          compliance-manager | components | composer | compute |
                          config | container | data-catalog |
                          database-migration | dataflow | datalineage |
                          dataplex | dataproc | datastore | datastream | deploy |
                          deployment-manager | design-center |
                          developer-connect | developer-knowledge | dns |
                          domains | edge-cache | edge-cloud | emulators |
                          endpoints | essential-contacts | eventarc | filestore |
                          firebase | firestore | functions | gemini |
                          healthcare | iam | iap | identity | ids |
                          infra-manager | kms | logging | looker | lustre |
                          managed-kafka | memcache | memorystore | metastore |
                          migration | ml | model-armor | monitoring | netapp |
                          network-connectivity | network-management |
                          network-security | network-services | notebooks |
                          observability | oracle-database | org-policies |
                          organizations | pam | parametermanager |
                          policy-intelligence | policy-troubleshoot | preview |
                          privateca | projects | publicca | pubsub | quotas |
                          recaptcha | recommender | redis | resource-manager |
                          run | scc | scheduler | secrets | service-directory |
                          service-extensions | service-health | services |
                          source | source-manager | spanner | sql | storage |
                          support | tasks | telco-automation | topic |
                          transcoder | transfer | vector-search | vmware |
                          workbench | workflows | workload-identity |
                          workspace-add-ons | workstations
  command may be          cheat-sheet | docker | feedback | help | info | init |
                          survey | version

For detailed information on this command and its flags, run:
  gcloud --help
```

```
student_03_e984a5d0fa06@cloudshell:~ (qwiklabs-gcp-04-4c4fa2b8f66c)$ gcloud config list
[accessibility]
screen_reader = True
[component_manager]
disable_update_check = True
[compute]
gce_metadata_read_timeout_sec = 30
region = asia-southeast1
zone = asia-southeast1-c
[core]
account = student-03-e984a5d0fa06@qwiklabs.net
disable_usage_reporting = False
project = qwiklabs-gcp-04-4c4fa2b8f66c
universe_domain = googleapis.com
[metrics]
environment = devshell
```

```
student_03_e984a5d0fa06@cloudshell:~ (qwiklabs-gcp-04-4c4fa2b8f66c)$ gcloud compute instances list
NAME: gcelab2
ZONE: asia-southeast1-c
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.148.0.2
EXTERNAL_IP: 34.87.107.33
STATUS: RUNNING
```

9. Task 2 — Filter command-line output

Step 12: List VM instances

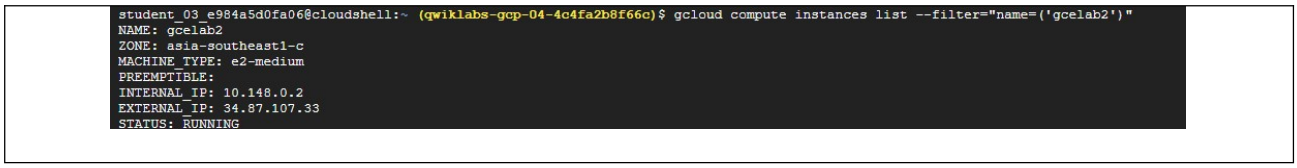
```
gcloud compute instances list
```

This lists Compute Engine VM instances available in the current project.

Step 13: Filter for gcelab2

```
gcloud compute instances list --filter="name=('gcelab2')"
```

The filter limits the displayed results to the instance named gcelab2.



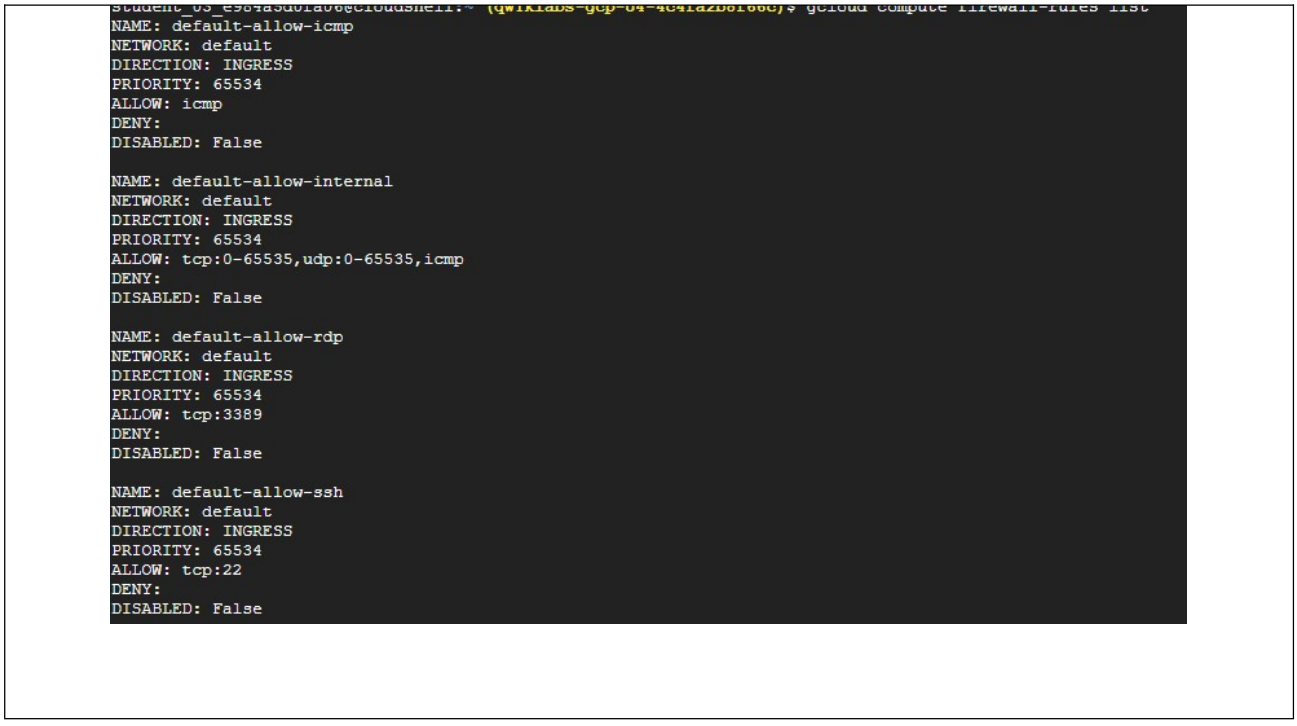
Step 14: List firewall rules

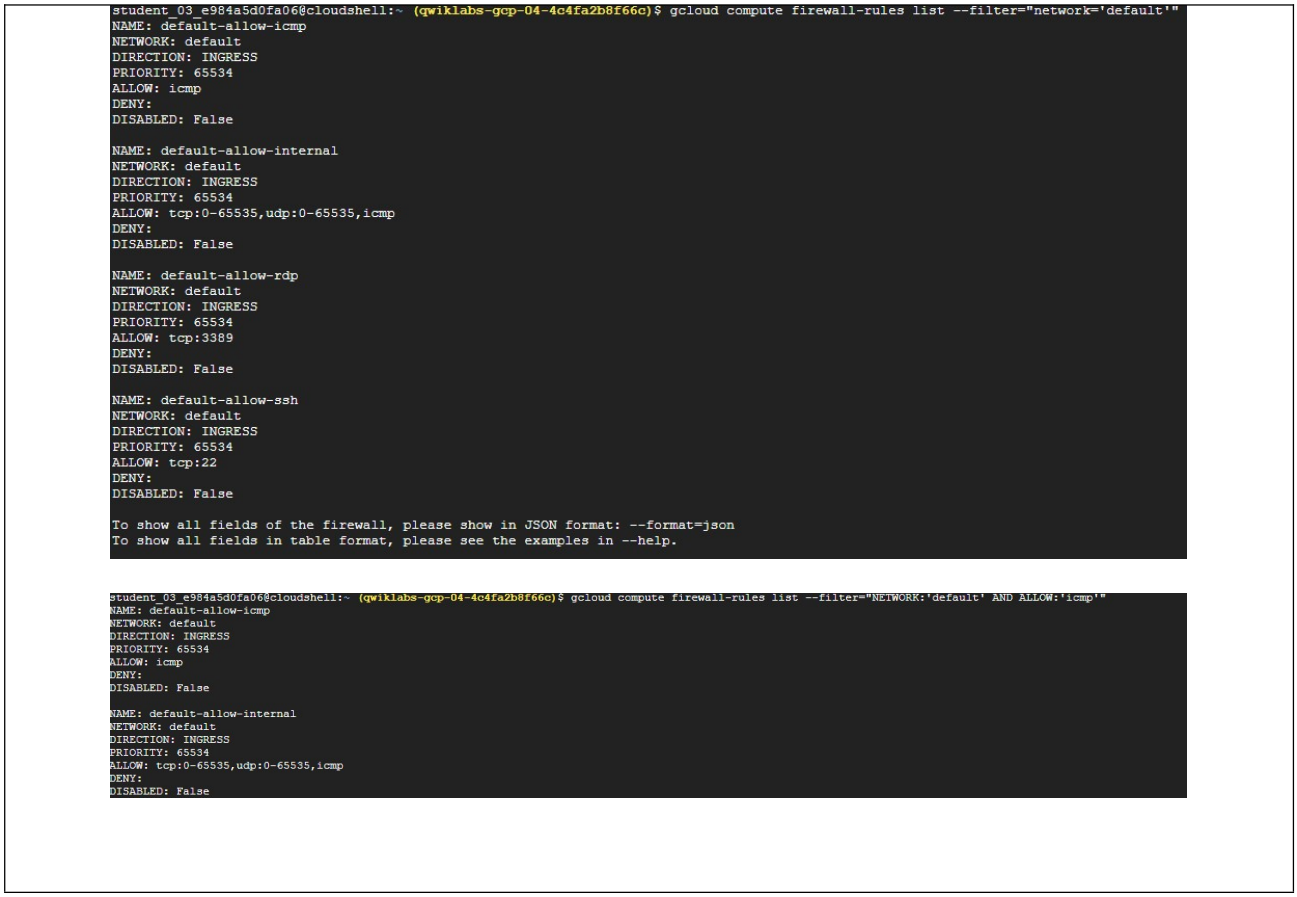
```
gcloud compute firewall-rules list

gcloud compute firewall-rules list --filter="network='default'"

gcloud compute firewall-rules list --filter="NETWORK:'default' AND ALLOW:'icmp'"
```

These commands list firewall rules, filter rules attached to the default network, and show rules whose allowed protocols include ICMP.



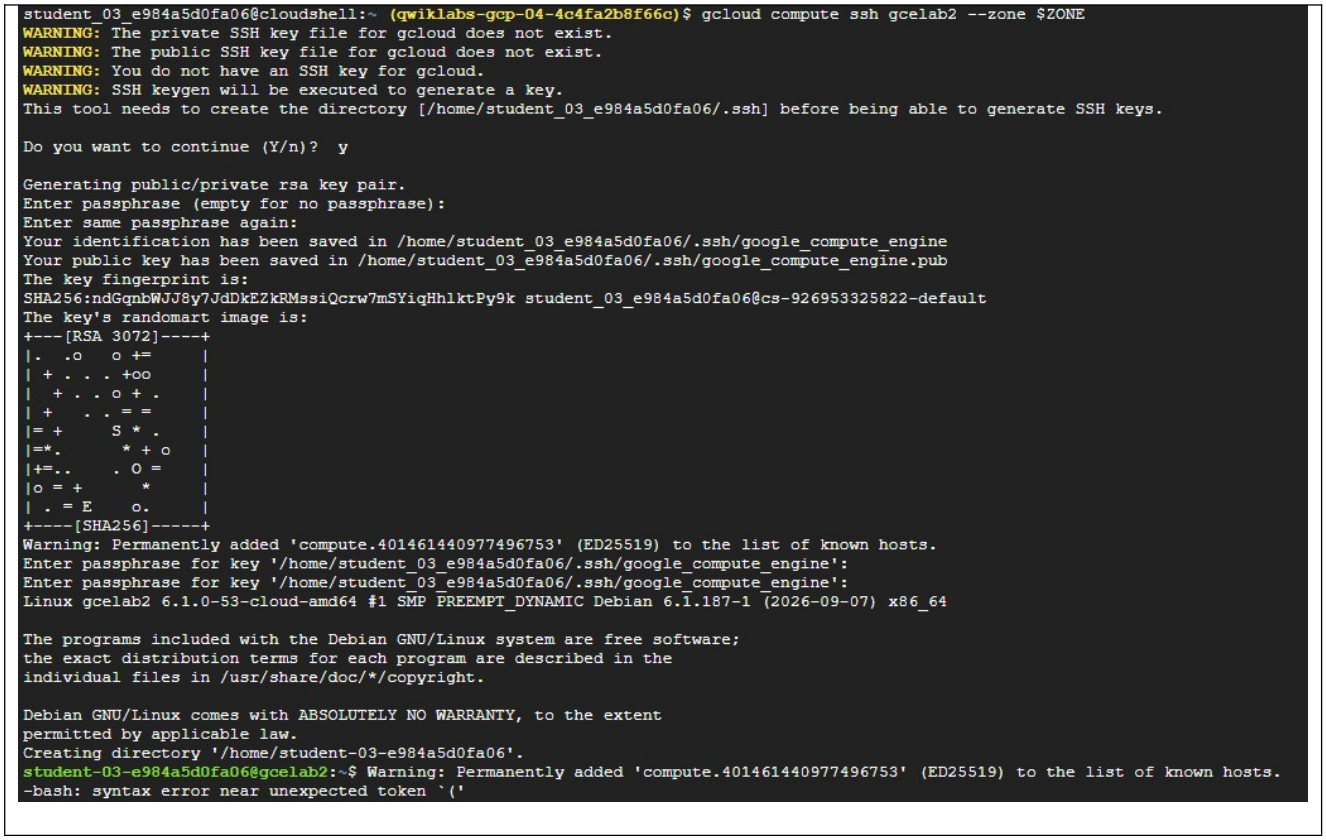


10. Task 3 — Connect to the VM using SSH

Step 15: Connect to gcelab2

```
gcloud compute ssh gcelab2 --zone $ZONE
```

On the first connection, gcloud may ask to create SSH keys. Type Y when asked to continue, then press Enter twice if you want to leave the key passphrase empty for this temporary lab. Wait until the shell prompt changes to indicate that you are connected to gcelab2.



Step 16: Install Nginx

```
sudo apt update && sudo apt install -y nginx
```

Wait for the package installation to complete.

Step 17: Exit the VM

```
exit
```

The terminal should return to the Cloud Shell project prompt.

11. Task 4 — Configure the firewall for HTTP

Step 18: Review firewall rules

```
gcloud compute firewall-rules list
```

The lab instructions note that the initial HTTP request may fail because TCP port 80 is not allowed by a suitable ingress firewall rule.

Step 19: Test the Nginx endpoint before adding the rule

```
curl http://$(gcloud compute instances list --filter=name:gcelab2 --format='value(EXTERNAL_IP)')
```

If the firewall has not been configured, the request may time out or fail. Continue with the next steps.

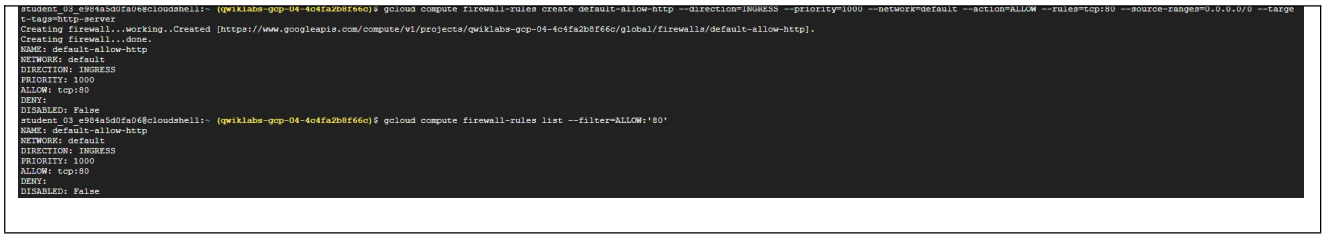
Step 20: Add HTTP and HTTPS network tags

```
gcloud compute instances add-tags gcelab2 --tags http-server,https-server
```

Step 21: Create an HTTP ingress firewall rule

```
gcloud compute firewall-rules create default-allow-http --direction=INGRESS --priority=1000 --
network=default --action=ALLOW --rules=tcp:80 --source-ranges=0.0.0.0/0 --target-tags=http-
server
```

This rule allows inbound TCP traffic on port 80 from any IPv4 source to instances carrying the http-server network tag on the default network. For production systems, restrict source ranges when possible and follow your organization's security policy.



Step 22: Verify the firewall rule

```
gcloud compute firewall-rules list --filter=ALLOW:'80'
```

Confirm that default-allow-http appears and that the allowed protocol is tcp:80.

Step 23: Test HTTP again

```
curl http://$(gcloud compute instances list --filter=name:gcelab2 --format='value(EXTERNAL_IP)')
```

The response should contain the default Nginx welcome page HTML. You can also open the VM external IP address in a browser using http:// if the lab allows it.



Return to the lab instructions and click Check my progress for "Update the firewall."

12. Task 5 — View system logs

Step 24: List available logs

```
gcloud logging logs list
```

This lists log names available in the current project. The exact names vary by project and enabled services.

Step 25: Filter logs related to Compute Engine

```
gcloud logging logs list --filter="compute"
```

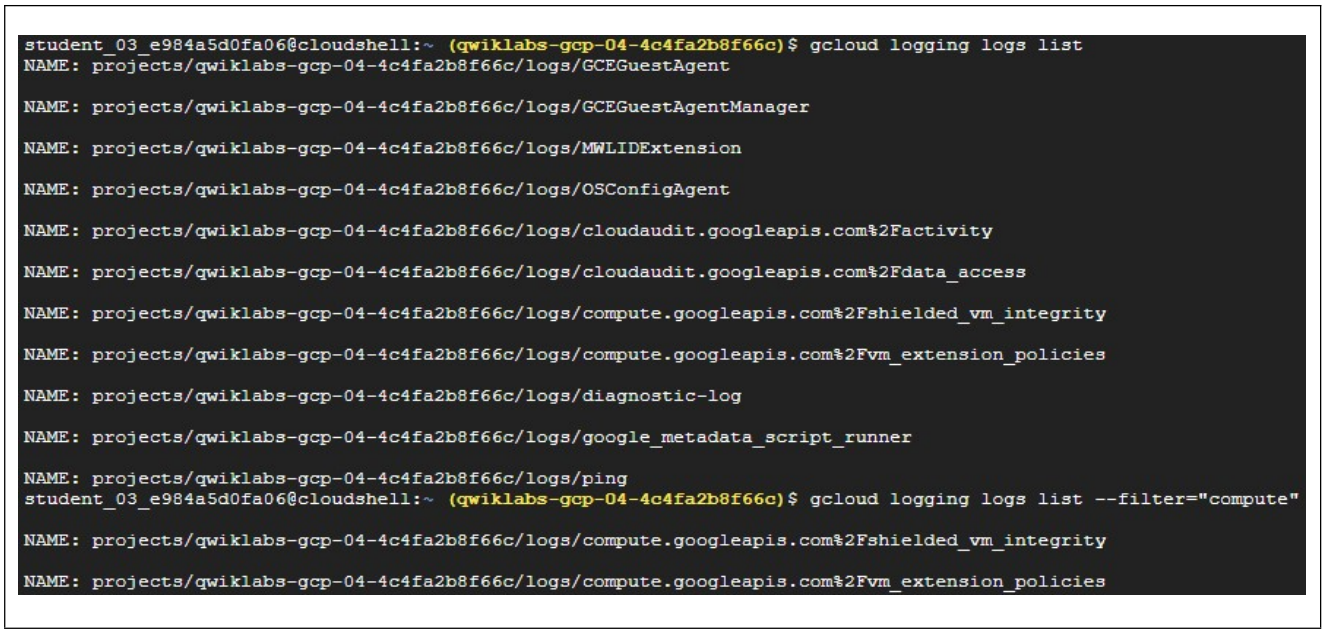
Step 26: Read recent VM logs

```
gcloud logging read "resource.type=gce_instance" --limit 5
```

Step 27: Read logs for gcelab2

```
gcloud logging read "resource.type=gce_instance AND labels.instance_name=gcelab2" --limit 5
```

Review timestamps, resource labels, and log messages. Results depend on which events have occurred and which logs are available.



13. Task 6 — Test your understanding

The supplied lab asks which three basic ways are used to interact with Google Cloud services and resources.

Correct choices from the supplied options:

- Command-line interface (CLI)
- Client libraries
- Cloud Console

GStreamer and GLib are not the three Google Cloud interaction methods listed in this lab question

14. Conclusion

In this lab, the gcloud CLI was used to configure a Google Cloud environment, create and access a Compute Engine VM, filter resource output, allow HTTP traffic through a firewall rule, test an Nginx web server, and inspect Cloud Logging. The screenshots should document the successful result of each major task.

End of document